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Batch no- 074

Topic: VPC basics and creation

- **What is VPC**?

VPC stands for virtual private cloud, this is isolated cloud structure provided by vendors it is use by owners to create private secure server using resources of their need.

Vpc is customisable user can customise this network according to the need

VPC is isolated network in which resources are private and secure

VPC is used for hosting web applications, creating secure servers.

- **Why we use VPC**?

Vpc provides isolation, high security for sensitive data, user has full control over the resources. VPC lets you run cloud resources in **your own isolated network**, just like a private data center.

- **Components of VPC**

1.**subnets**: We use subnets to divide a big network into smaller, manageable, and secure networks.

- **Private subnets**: subnets who don't have internet access , which use for more sensitive data.
- **Public subnets**: subnets who has internet access via IGW.

2.**route tables**: A route table defines how network traffic is routed within a VPC and to external networks.

3.**IGW(internet gateway)**: This provides Internet to our private VPC via internet gateway so we can update or access it publicly.

4.**NAT gateway**: To convert our private VPC to public or to access the internet we use NAT gateway.

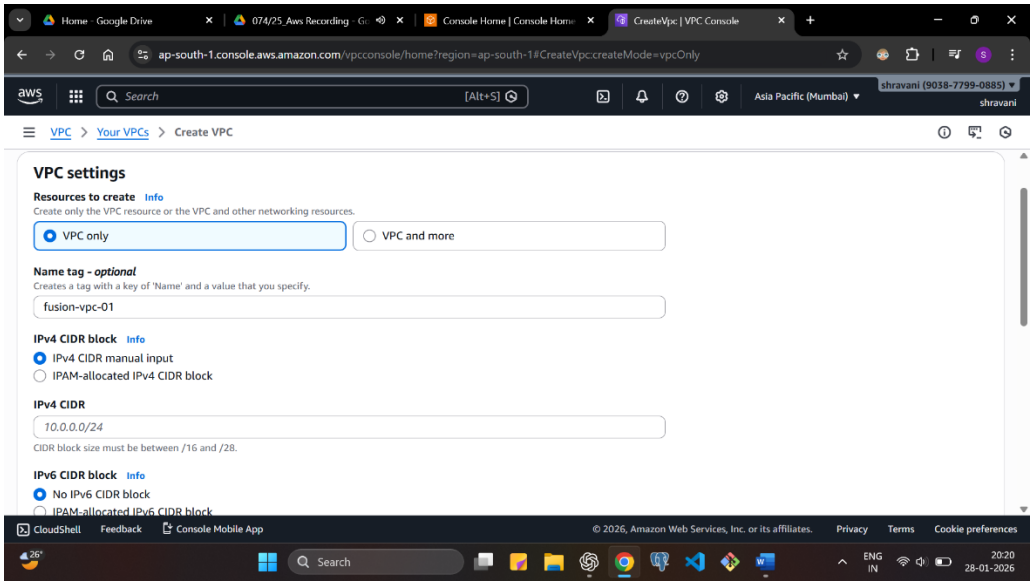
- **CIDR**: class less inter domain routing which use to restrict IP

Example: vpc /16, subnets /24.

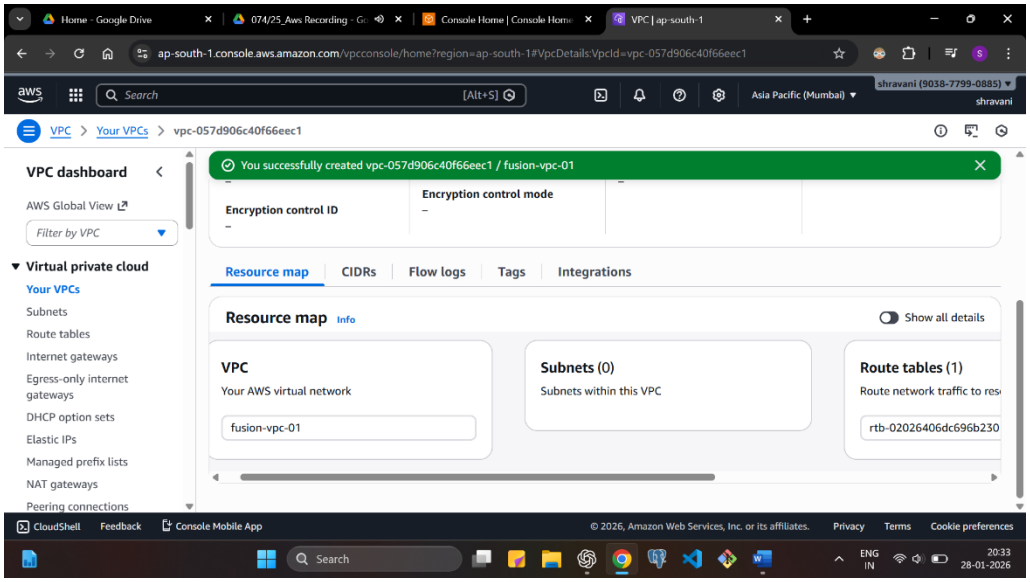
- **Steps to create VPC:**

You can create 5 VPC in one region.

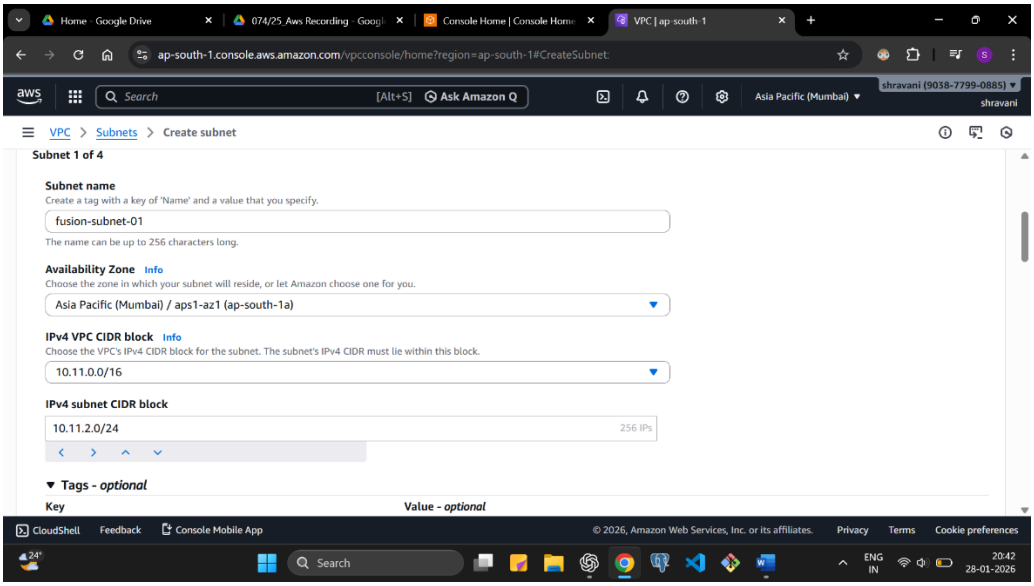
1. To create VPC go to AWS console open VPC service go to your VPC default VPC is already there. Now to create our own vpc click on create VPC select VPC only give name and CIDR range then click on create



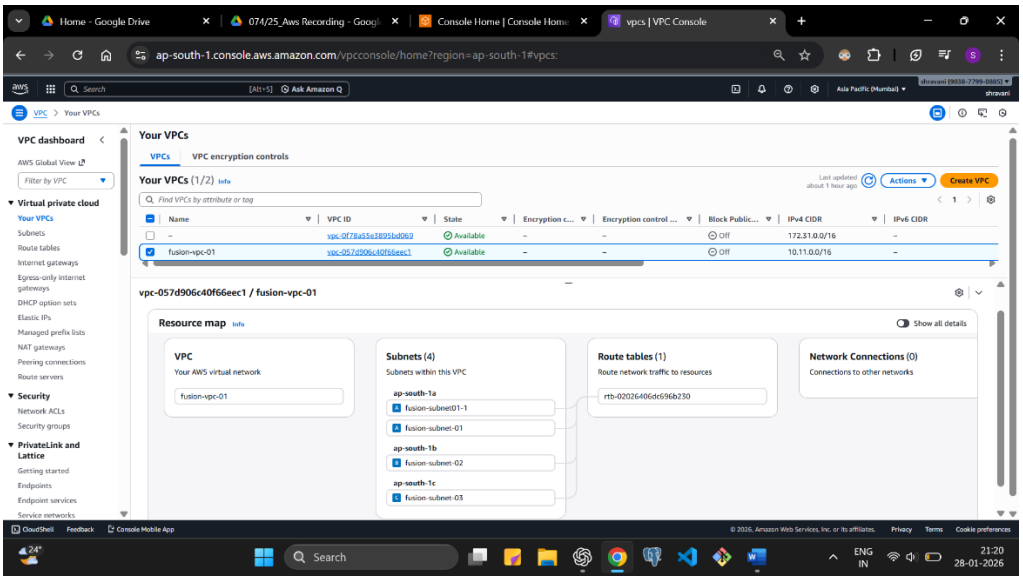
2. After creating vpc it will show you route tables also have been created.



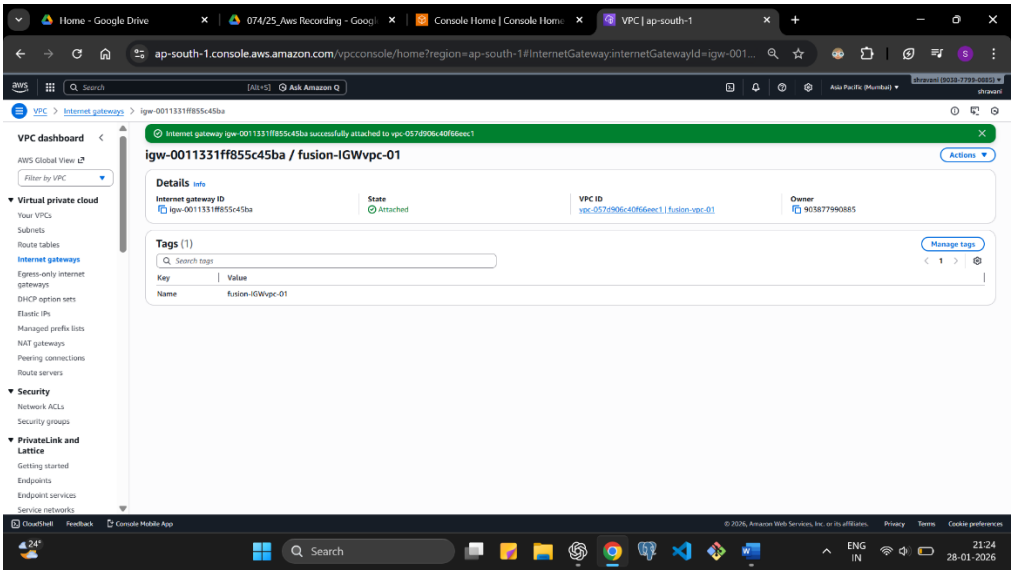
3. Now to create subnet go to subnet select our VPC give name to subnet select availability zone, you can create multiple subnets under 3 AZ (min subnet in region)



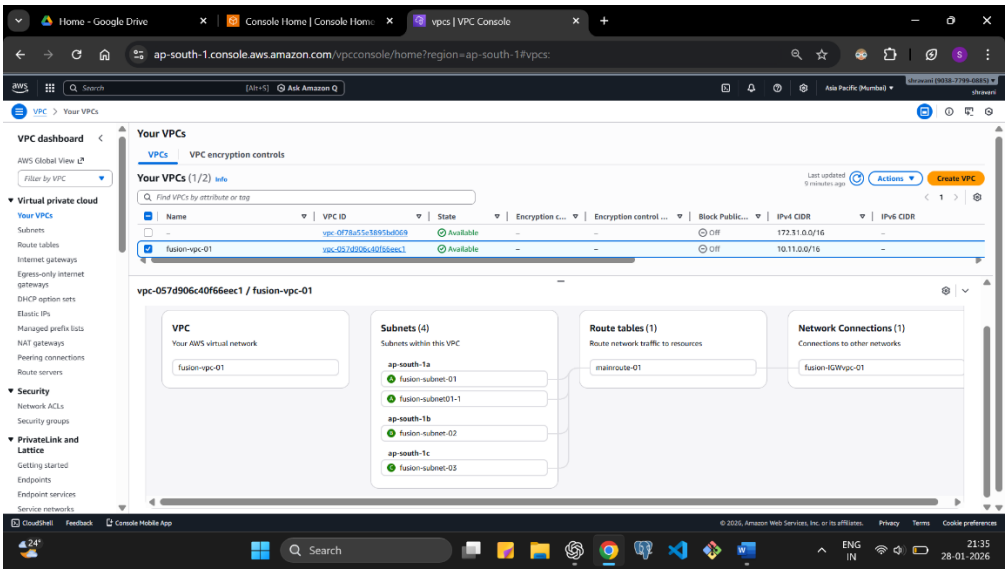
4. After creating subnets , to create IGW (internet gateway) click in create internet gateway give name to IGW and after creation attached it to our created VPC .



5. To connect it route table go to route table click on created route table and rename it after go to routes and go to edit routes click add new routes



6. Now our own VPC has been created now we can use this.



7. After connecting don't forget to delete your created VPC and it automatically deletes all created subnets and IG.